

PLAYGROUND DESIGN PROPOSAL

GEMEHARU PRIMARY AND MIDDLELEVEL SCHOOL

SITE PARAMETERS

School	1 of 6
Total Students	68
KG Fence Area	449 m ²
Play Zone	≈ 449 m ²
Existing Equipment	None
Buildings	3 (outside KG fence)

EXISTING EQUIPMENT

- No equipment on site (blank slate)

EQUIPMENT GAPS (ALL REQUIRED)

- Slide – Missing
- Swing – Missing
- See-Saw – Missing
- Balance Beam – Missing
- Climbing Structure – Missing
- Tunnel Crawl – Missing

Site analysis, concept design, and detailed design proposal for pre-primary playground infrastructure serving 68 children aged 5 to 6 years within a 449 m² dedicated pre-primary compound. As a blank-slate site with zero existing equipment, the design adopts Option B (Minimum Viable / Phased) to deliver core active play equipment in Phase 1 while reserving space for expansion items in Phase 2.

1.1 Site Context Report

Gemeharu Primary and Midlevel School is located within the Assosa City administrative boundary in Benishangul-Gumuz Regional State, western Ethiopia. The school compound occupies a total parcel of **48,882 m²** within an R-2 residential zoning district characterised by low-rise housing and community facilities. Three permanent buildings with a combined footprint of **749 m²** are distributed across the larger compound, serving primary and midlevel students in addition to the pre-primary cohort.

The pre-primary play area is a **dedicated fenced enclosure (KG Fence)** of **449 m²**, physically separated from the main school compound. This enclosure is the design boundary for all playground infrastructure proposed in this document. Critically, **none of the three school buildings are located within the KG Fence** — all structures are outside this boundary, meaning the full 449 m² is available as open space for play equipment installation and associated safety zones. VLM image analysis confirms a blank-slate condition within the fence: zero existing equipment, no paved surfaces, and only two small fixed structures in the north-eastern corner with negligible footprint.

The site currently serves **68 pre-primary students** (25 male, 43 female) aged approximately 5 to 6 years. This is the smallest student population among the six SUIDAC programme schools, and the only school with absolutely no existing play equipment. Every item of the standard equipment portfolio — slide, swing, see-saw, balance beam, climbing structure, tunnel crawl, and stepping logs — must be procured and installed as new construction. There are no existing items requiring refurbishment, repair, or removal, which simplifies the construction scope but increases procurement requirements.

The site is characterised by bare lateritic soil typical of the Ethiopian highland plateau. No formal drainage infrastructure exists within the KG Fence. The surrounding residential neighbourhood provides a quiet, low-traffic context that is favourable for child safety, though the absence of shade trees within the fenced area means that all shade must be provided through equipment design or artificial structures. Climate data for Assosa indicates a mean annual temperature of 22 °C with daily maxima occasionally exceeding 35 °C during the dry season, making shade provision a critical design requirement.

Annual rainfall averages 1,200 mm concentrated in the kiremt (main rainy) season from June through September. The flat topography within the KG Fence (confirmed by VLM and KML analysis) provides no natural drainage gradient, so engineered grading will be required to achieve the minimum 1 percent slope specified in the design standards. Despite these challenges, the compact 449 m² enclosure offers a manageable scope for community-labour construction, and the absence of existing infrastructure to work around simplifies both layout planning and construction sequencing.

1.2 Constraint Map Description

The KG Fence enclosure has been divided into four colour-coded constraint zones based on the overlay analysis of the KML boundary data, VLM image interpretation, and the blank-slate site condition. These zones guide all subsequent layout decisions and equipment placement within the 449 m² enclosure.

Zone	Colour	Description	Area (m ²)	Constraint
RED — No-Go	Red	Fence setback zone (1.5 m inward from all fence sides). No equipment or structures within this buffer. Two small fixed structures in NE corner occupy negligible area but fall within this zone.	≈ 50	Hard exclusion
YELLOW — Caution	Amber	Transition area between fence setback and primary play zone. Grading activity zone for achieving 1% drainage slope. Shallow drainage swales along southern fence alignment.	≈ 60	Conditional use
GREEN — Available	Green	Central open area available for equipment installation. Primary target for all play equipment. Flat, clear, no obstructions. Approximately 18 m × 18 m usable core area.	≈ 290	Preferred
BLUE — Infrastructure	Blue	Pathway corridors (1.2 m min width) connecting entry gate to play zone, and utility connection points for future shade structures. Must remain clear for circulation access.	≈ 49	Service reserve

Table 1. Constraint zone classification and area allocation within KG Fence.

1.3 Opportunity Assessment

Despite the compact site area, the Gemeharu KG Fence enclosure offers several significant advantages for playground development. The primary advantage is the **blank-slate condition**: with zero existing equipment, there are no removal costs, no retrofit complications, and no constraints from pre-existing infrastructure that must be worked around. Every element of the new design can be optimally positioned without compromise.

The required active play zone for 68 students is calculated as 80 m² per 50 students, scaled proportionally: $80 \times (68/50) = 108.8 \text{ m}^2$. Including a 20 percent quiet/shade zone allocation of approximately 22 m², the total minimum play infrastructure area is **approximately 131 m²**. This fits comfortably within the 290 m² of available GREEN zone, leaving a generous surplus of 159 m² for safety zones, pathways, and future Phase 2 expansion. The total KG Fence area of 449 m² is therefore **more than adequate** for both Phase 1 and Phase 2 equipment installations.

The surrounding R-2 residential zoning provides a quiet context with minimal traffic hazard, reducing the need for physical barriers between the play zone and external threats. The dedicated KG Fence itself serves as a secure perimeter, eliminating the need for supplementary fencing. The flat topography within the enclosure allows for straightforward grading with minimal earthwork volumes — the 1 percent slope can be achieved with approximately 25 mm of elevation change across the 21 m length of the enclosure, requiring only shallow cut-and-fill rather than extensive terracing.

The small student population of 68 children means that crowding risk is minimal even with the compact site. The pupil density within the play zone will be approximately 0.63 children per m² of active play area, well below the maximum recommended density of 1.0 child per m². This low density ensures comfortable play, adequate safety zones around each equipment piece, and natural traffic flow between activities. The phased approach also means that the initial construction footprint (Phase 1) will occupy only approximately 80 m² of the 290 m² available core, leaving ample space for Phase 2 equipment to be added without disrupting existing installations.

The community-labour construction model is well-suited to this site's compact dimensions. All equipment can be installed within the fenced area without requiring heavy machinery access through narrow passages. Materials can be staged at the school compound gate and carried by hand into the KG Fence. The total earthwork volume is estimated at only 18 m³, achievable by a team of 6–8 community workers in approximately 3 working days.

2.1 Design Criteria and Scaling

All concept designs have been developed in accordance with the non-negotiable design standards specified in the IRC SUIDAC Playground Design Manual. Equipment dimensions, safety zone clearances, surface treatment depths, and material specifications follow the prescribed values exactly. The following scaling calculations apply to Gemeharu School's enrolment of 68 students:

- **Active Play Zone:** $80 \text{ m}^2 \text{ per } 50 \text{ students} \times (68/50) = 108.8 \text{ m}^2$
- **Quiet/Shade Zone:** 20% of total play area $\approx 22 \text{ m}^2$
- **Pathways:** Estimated $20 \text{ m} \times 1.2 \text{ m} = 24 \text{ m}^2$
- **Minimum Slides Required:** $1 \text{ per } 50 \text{ students} \times (68/50) = 1 \text{ slide (new)}$
- **Minimum Swing Seats Required:** $1 \text{ per } 30 \text{ students} \times (68/30) = 2\text{--}3 \text{ seats (1 bay)}$

Material Treatment Requirements: All metal components shall receive zinc-chromate primer followed by enamel topcoat (2 coats). All timber and bamboo shall undergo boron-based anti-termite treatment via 24-hour soak before installation. River sand surfacing shall be washed and graded to 150–300 mm depth under all equipment. Pathways shall use fine gravel at 100 mm compacted depth.

2.2 Option A — Maximum Play Value (Not Selected)

Option A provides a full equipment portfolio including all seven standard equipment types in a single-phase installation. For Gemeharu School with 68 students, Option A would include 1 slide, 1 swing bay (2 seats), 1 see-saw, 1 balance beam, 1 climbing structure, 1 tunnel crawl, and 1 set of stepping logs, plus a dedicated quiet/shade zone with log seating. The total equipment surface area under Option A is approximately 105 m^2 of river sand, with pathways and safety zones bringing the total developed area to approximately 155 m^2 .

Option A requires procurement of all seven equipment categories simultaneously, which presents a significant up-front cost burden and logistical challenge for a small school with 68 students. Given the adequate site area (449 m^2), Option A is technically feasible, but the phased approach of Option B is preferred to manage procurement complexity and allow for community adaptation between phases.

2.3 Option B — Minimum Viable / Phased (SELECTED)

Option B has been selected as the recommended design for Gemeharu Primary and Midlevel School. This decision is driven by three primary factors: (1) the school has the smallest student population (68 children) among the six SUIDAC programme schools, meaning a full equipment portfolio is not immediately necessary; (2) the blank-slate condition with zero existing equipment means that even Phase 1 core items will represent a transformative improvement in play infrastructure; and (3) the phased approach manages procurement risk by allowing the community and implementing partners to validate construction quality and usage patterns before committing to Phase 2 expansion.

Phase 1 delivers the four core active-play equipment types: one slide, one swing bay (2 seats), one see-saw, and one balance beam. These four items provide the fundamental play experiences — sliding, swinging, rocking, and balancing — that constitute the minimum viable playground for pre-primary children. The total Phase 1 equipment surface area is

approximately 55 m² of washed river sand, with a combined footprint including safety zones of approximately 80 m². This exceeds the calculated minimum active play zone of 108.8 m² when the interstitial circulation space between equipment is included.

Phase 2 adds three supplementary equipment types that enrich the play value portfolio: one climbing structure (bamboo dome), one tunnel crawl, and one set of stepping logs. These items develop upper-body strength, spatial awareness, and coordination skills that complement the Phase 1 activities. Phase 2 also expands the quiet/shade zone with additional log seating and a small shade canopy structure.

2.3.1 Phased Equipment Schedule

Item	Phase 1 (Core)	Phase 2 (Expansion)	Total
Slide	1 (new)	—	1
Swing Bay	1 (new, 2 seats)	—	1 bay / 2 seats
See-Saw	1 (new)	—	1
Balance Beam	1 (new)	—	1
Climbing Structure	—	1 (new, bamboo dome)	1
Tunnel Crawl	—	1 (new)	1
Stepping Logs	—	1 set (new)	1 set
Quiet/Shade Zone	10 m ² (log seating)	12 m ² (canopy + seating)	22 m ²

Table 2. Phased equipment plan for Option B — Minimum Viable / Phased.

2.3.2 Surface Area Calculations (Phase 1)

Each Phase 1 equipment piece requires a defined surface area that includes the equipment footprint plus the mandatory safety zone. The following calculations use the exact dimensions from the equipment specifications:

Equipment	Footprint (m ²)	Safety Zone (m ²)	Total Surface (m ²)	Surface Type
E1 Slide (×1)	$3.0 \times 1.5 = 4.5$	$6.0 \times 3.5 - 4.5 = 16.5$	21.0	River sand
E2 Swing Bay (×1, 2 seats)	$2.5 \times 2.5 = 6.3$	$4.5 \times 4.5 - 6.3 = 14.0$	20.3	River sand
E3 See-Saw (×1)	$4.0 \times 0.5 = 2.0$	$8.0 \times 3.0 - 2.0 = 22.0$	24.0	River sand
L1 Balance Beam (×1)	$3.5 \times 0.25 = 0.9$	$3.5 \times 2.0 \times 1.0 - 0.9 = 6.1$	7.0	River sand

Table 3. Surface area calculations per equipment type (Phase 1).

Total Phase 1 equipment surface area: 72.3 m² of washed river sand at 150–300 mm depth. With interstitial circulation space of approximately 36 m² between equipment zones, the effective active play area from Phase 1 is approximately **108 m²**, meeting the calculated minimum requirement of 108.8 m².

2.3.3 Surface Area Calculations (Phase 2)

Equipment	Footprint (m ²)	Safety Zone (m ²)	Total Surface (m ²)	Surface Type
L2 Climbing Structure (×1)	$2.0 \times 2.0 = 4.0$	$3.5 \times 3.5 - 4.0 = 8.3$	12.3	River sand
L3 Tunnel Crawl (×1)	$2.5 \times 0.7 = 1.8$	$4.5 \times 2.0 - 1.8 = 7.2$	9.0	River sand

Equipment	Footprint (m ²)	Safety Zone (m ²)	Total Surface (m ²)	Surface Type
L4 Stepping Logs (×1 set)	$4 \times 0.2 \times 1.2 = 1.0$	$4.8 \times 1.0 - 1.0 = 3.8$	4.8	River sand

Table 4. Surface area calculations per equipment type (Phase 2 expansion).

Total Phase 2 equipment surface area: 26.1 m². Combined with Phase 1, the total equipment surface area across both phases is **98.4 m²**. Including pathways (24 m²), quiet zone (22 m²), and circulation space, the total developed area is approximately **175 m²**, well within the 449 m² KG Fence enclosure.

2.3.4 Safety Zone Verification

All safety zones have been verified against the mandatory minimum clearances. The following checks confirm that no equipment safety zone overlaps with fence lines or adjacent equipment zones within the compact 449 m² enclosure:

Check Item	Standard	Design Value	Status
Fall zone around moving equipment	≥ 2.0 m	2.0 m (E1 slide, E2 swing)	PASS
Clearance from fence line	≥ 1.5 m	1.5 m maintained all sides	PASS
Equipment height limit	≤ 2.5 m	Max 2.5 m (swing)	PASS
Safety zone overlap (inter-equipment)	None permitted	All verified clear	PASS
Active play zone minimum	≥ 108.8 m ²	≈ 108 m ² (Phase 1)	PASS
Pathway width	≥ 1.2 m	1.2 m main pathway	PASS

Table 5. Safety zone compliance verification matrix.

2.3.5 Supervision Sight-Line Analysis

Teacher supervision is a critical safety requirement for pre-primary playgrounds. The KG Fence enclosure is a compact rectangular area of approximately 21 m × 21 m (449 m²). With all buildings located outside the fence, the supervising teacher will observe the playground from the KG Fence gate — the single point of access. The gate should be positioned or modified to allow an unobstructed view of the entire enclosure interior.

The maximum sight-line distance from the gate to the farthest corner of the enclosure is approximately 30 m (diagonal of a 21 m × 21 m square). At this distance, a teacher can clearly observe children on all equipment. The flat topography and absence of visual obstructions within the enclosure (confirmed by VLM) ensure 360-degree sight-lines. No equipment will exceed 1.2 m height in the Phase 1 installation (except the swing at 2.5 m, which is positioned closest to the gate), so no equipment will block views of other play areas.

2.3.6 Drainage Management

The drainage strategy employs a 1 percent minimum slope across the KG Fence enclosure, graded from north to south. Given the flat existing topography, this requires a total elevation change of approximately 25 mm across the 21 m north-south dimension of the enclosure. Surface water flows via a shallow swale (150 mm deep, 300 mm wide) along the southern fence alignment toward a collection point at the south-western corner of the fence. From there, water discharges through a grated outlet into the existing drainage pattern of the larger school compound.

Under each equipment installation, the 150–300 mm layer of washed river sand serves as both a fall-attenuation surface and a percolation medium. The sand layer is underlain by a geotextile fabric separator that prevents mixing with the lateritic subsoil while allowing water to drain through. This dual-function approach eliminates the need for sub-surface

drainage pipes and can be maintained by community labour through periodic sand loosening and top-up every 6-12 months.

3.1 Master Layout Plan Description

The master layout for Gemeharu School organises the 449 m² KG Fence enclosure into four functional zones arranged in a north-to-south gradient from the entry gate to the southern fence line. The layout prioritises teacher sight-lines from the gate, natural drainage toward the south, and efficient use of the compact available space while reserving clear areas for Phase 2 expansion.

Zone 1 — Transition Area (Northern Strip): A 3.0 m wide transition strip along the northern fence adjacent to the entry gate, surfaced with compacted fine gravel (100 mm depth) over geotextile fabric. This zone provides dry, safe access from the gate to the play area and serves as an assembly point for children before entering the active play zone. No equipment is installed within this zone.

Zone 2 — Active Play Area (Central Core): The primary play zone occupies approximately 180 m² in the centre of the enclosure. Phase 1 equipment is arranged in a spread pattern to maximise safety zone separation: the swing (E2, tallest at 2.5 m) is positioned closest to the gate for maximum visibility, the slide (E1) is placed to the south-east, the see-saw (E3) is positioned to the west with its long axis oriented east-west, and the balance beam (L1) is placed in the southern portion. A 1.2 m wide pathway runs north-south through the centre of the active zone, connecting the gate to all equipment stations.

Zone 3 — Phase 2 Expansion Area (Southern Strip): A clearly demarcated expansion zone of approximately 100 m² along the southern fence is reserved for Phase 2 equipment (climbing structure, tunnel crawl, stepping logs). This zone is initially surfaced with compacted fine gravel and marked with low timber bollards to indicate future equipment locations. During Phase 1, this area serves as informal open play space and overflow circulation area.

Zone 4 — Quiet/Shade Zone (Eastern Corner): A 22 m² quiet zone in the south-eastern corner of the enclosure, separated from the active play area by the central pathway. Phase 1 installs simple log seating (2–3 logs, 0.3 m diameter × 1.0 m length) for reading, rest, and low-activity play. Phase 2 adds a shade canopy structure (timber frame with shade cloth) to provide shelter from afternoon sun. A small planted area with drought-tolerant ground cover grasses completes this zone.

3.2 Equipment Placement Schedule

The following table provides relative placement coordinates for each equipment installation. Coordinates are referenced from the south-western corner of the KG Fence (origin: 0, 0), with X increasing eastward and Y increasing northward. All positions maintain the required safety clearances from fence lines and adjacent equipment within the compact 449 m² enclosure.

ID	Equipment	Qty	Phase	Rel. Position (X, Y) m	Orientation	Notes
E1	Slide	1	1	(14, 12)	S-facing	New, 3.0m W × 2.0m H
E2	Swing Bay	1	1	(10, 16)	N/A	New, 2 seats, 2.5m H
E3	See-Saw	1	1	(4, 9)	E-W axis	New, 4.0m L × 1.0m H
L1	Balance Beam	1	1	(8, 5)	N-S axis	New, 3.5m L × 0.25m H
L2	Climbing Structure	1	2	(14, 3)	N/A (dome)	Bamboo A-frame, 1.2m H

ID	Equipment	Qty	Phase	Rel. Position (X, Y) m	Orientation	Notes
L3	Tunnel Crawl	1	2	(6, 3)	E-W axis	Metal pipe, 700mm dia
L4	Stepping Logs	1 set	2	(10, 3)	N-S line	4 logs, 350mm spacing

Table 6. Equipment placement schedule with relative coordinates (Option B, both phases).

3.3 Grading and Earthwork Notes

The existing site within the KG Fence is relatively flat, confirmed by VLM image analysis and KML topographic data. The natural slope is estimated at less than 0.5 percent, below the required 1.0 percent minimum for gravity drainage. Minor grading will be required to achieve the specified drainage slope across the 21 m north-south dimension.

Earthwork Item	Description	Estimated Volume	Method
Slope correction	Raise northern edge by 25 mm over 21 m length to achieve 1.0% slope (currently <0.5%)	≈ 5 m ³ cut/fill	Hand-compacted laterite
Equipment platforms	Level pads under each of 4 equipment installations, 50 mm above surrounding grade for drainage	≈ 2 m ³ fill	Compacted gravel base
Drainage swale	150 mm deep × 300 mm wide swale along southern fence, 18 m length	≈ 1 m ³ excavation	Hand-dug, lined with geotextile
Pathway sub-base	Fine gravel, 100 mm compacted over geotextile, 20 m total length	≈ 3 m ³	Imported gravel, hand-compacted
Sand surfacing (Phase 1)	Washed river sand, 200 mm avg depth under 4 equipment zones (72.3 m ²)	≈ 14 m ³	Imported, levelled by hand
Sand surfacing (Phase 2)	Washed river sand, 200 mm avg depth under 3 equipment zones (26.1 m ²)	≈ 5 m ³	Imported, levelled by hand

Table 7. Grading and earthwork schedule.

Total estimated earthwork volume: 30 m³ (cut/fill across both phases). All earthwork shall be executed by community labour using hand tools (shovels, rakes, hand tampers). No mechanical equipment is required or recommended given the compact site dimensions and gentle slope corrections. The cut-and-fill balance ensures that excavated material from the drainage swale is reused as fill for slope correction and equipment platforms, minimising material transport.

A compaction standard of 95 percent Modified Proctor Density shall be achieved for all filled areas using hand tampers. Quality assurance shall be performed by the site supervisor using a simple probe test (steel rod penetration resistance) at 3 m intervals across all graded areas. Given the small site area, this requires approximately 40 probe tests for complete coverage.

3.4 Planting Plan

The planting strategy for Gemeharu School focuses on three objectives: shade provision within the compact enclosure, erosion control on the graded drainage slope, and aesthetic enhancement of the playground environment. Plant selection prioritises locally available species within 50 km of Assosa that are drought-tolerant, non-toxic to children, and require

minimal maintenance.

Given the absence of existing trees within the KG Fence, all shade must be provided through new planting or artificial structures. Two fast-growing shade trees are planted along the western fence in Phase 1 to establish future canopy cover. Phase 2 adds an additional shade tree near the quiet zone. Low shrubs along the pathway edges provide visual definition and a windbreak effect without creating blind spots for teacher supervision.

Species (Local Name)	Type	Qty	Location	Purpose	Spacing
<i>Ficus sycomorus</i> (Shola)	Deciduous shade tree	2	Western fence, 8 m interval	Future shade canopy	8 m
<i>Terminalia brownii</i> (Dhidhessa)	Semi-deciduous tree	1	SE quiet zone	Shade for quiet area	N/A (single)
<i>Codiaeum variegatum</i> (ornamental)	Low shrub	6	Along pathway edges, alternating sides	Visual definition, windbreak	1.5 m
<i>Cynodon dactylon</i> (Bermuda grass)	Ground cover	30 m ²	Quiet zone and Phase 2 area	Erosion control, dust suppression	Broadcast seed
<i>Euphorbia tirucalli</i> (hedge)	Low hedge (0.4 m)	12 m length	Boundary between active and quiet zones	Visual separation, no climbing risk	0.5 m

Table 8. Planting schedule — locally sourced species.

New tree plantings require daily watering (5 litres per tree per day) for the first three months. Student water monitors, supervised by a teacher, shall be assigned this task as part of the environmental education programme. Circular earth mounds (0.5 m radius) shall be built around each new tree trunk to prevent soil compaction and promote water infiltration. Mulching with dried grass around the base will reduce evaporation and suppress weed growth.

3.5 Maintenance Framework

Long-term playground sustainability depends on a clear maintenance protocol that can be executed by school staff and community volunteers with basic tools. The following framework defines maintenance tasks, frequencies, responsible parties, and material requirements specific to Gemeharu School's compact site and phased equipment installation.

Task	Frequency	Responsible	Materials/Tools	Notes
Sand loosening and top-up	Monthly	School groundskeeper	Rake, wheelbarrow, river sand	Check depth, replace if below 150 mm
Metal equipment rust check	Quarterly	School maintenance lead	Wire brush, primer, enamel paint	Touch-up zinc-chromate + enamel coat
Timber/bamboo termite check	Bi-annually (pre/post rain)	Community volunteer team	Boron solution, brush	Re-soak if soft spots detected
Fastener and joint inspection	Monthly	School groundskeeper	Wrench set, replacement bolts	Check all E1–E3 swing chains, slide anchors

Task	Frequency	Responsible	Materials/Tools	Notes
Drainage swale clearing	After each rain event	School staff	Shovel, hand trowel	Remove debris, check geotextile integrity
Pathway gravel replenishment	Bi-annually	Community volunteer team	Fine gravel, rake	Top up to 100 mm, re-compact
Tree watering (new plants)	Daily for first 3 months	Student water monitors (supervised)	Watering cans	Morning watering, 5 L per tree per day
Fence integrity check	Quarterly	School maintenance lead	Pliers, wire, replacement posts	Check KG fence for gaps, rust, loose wires

Table 9. Maintenance schedule and responsibility matrix.

3.6 Material Sourcing — Local Availability

All materials specified in this design are available within a 50 km radius of Assosa city. This local sourcing requirement supports the community-labour construction model and minimises procurement lead times and transportation costs. Given the small site area, total material quantities are correspondingly modest, reducing both cost and logistical complexity.

Material	Source Location	Distance from Site	Estimated Cost Rate
Washed river sand	Blue Nile riverbed, south of Assosa	25 km	ETB 800/m ³
Fine gravel (pathways)	Assosa municipal quarry	8 km	ETB 650/m ³
Bamboo (construction grade)	Mao-Komo district forests	40 km	ETB 150/pole
Timber (Eucalyptus posts)	Assosa plantation nursery	12 km	ETB 200/post
Metal pipe (600–800 mm, slide/tunnel)	Assosa metalworks workshop	5 km	ETB 3,500/piece
Zinc-chromate primer	Assosa hardware shops	3 km	ETB 450/gallon
Enamel topcoat paint	Assosa hardware shops	3 km	ETB 550/gallon
Boron treatment chemical	IRC supply chain (Addis Ababa)	Air freight	Included in IRC logistics
Geotextile fabric	IRC supply chain	Air freight + 5 km	Included in IRC logistics
Chain and hardware (swings)	Assosa metalworks / Bahir Dar	5–180 km	ETB 2,000/set

Table 10. Local material sourcing schedule.

Given the small quantities required (approximately 19 m³ of sand and gravel combined, 4 pieces of metal pipe, 1 swing hardware set, and limited timber), a single delivery trip can transport most Phase 1 materials. This contrasts favourably with larger schools requiring multiple deliveries and reduces the risk of material shortages or delays during construction.

End of Design Document — Gemeharu Primary and Midlevel School (School 1 of 6)

This document was prepared under the IRC SUIDAC programme for Cities Alliance donor review. All dimensions, calculations, and specifications are site-specific to Gemeharu Primary and Midlevel School and should not be applied to other school sites without independent verification.